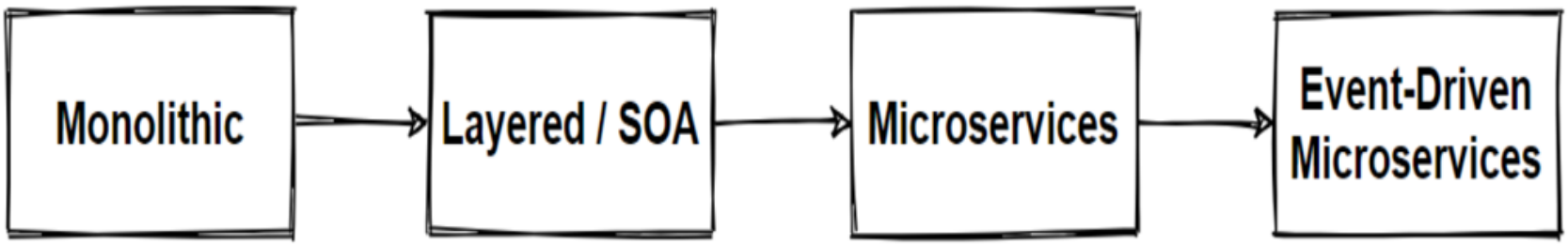
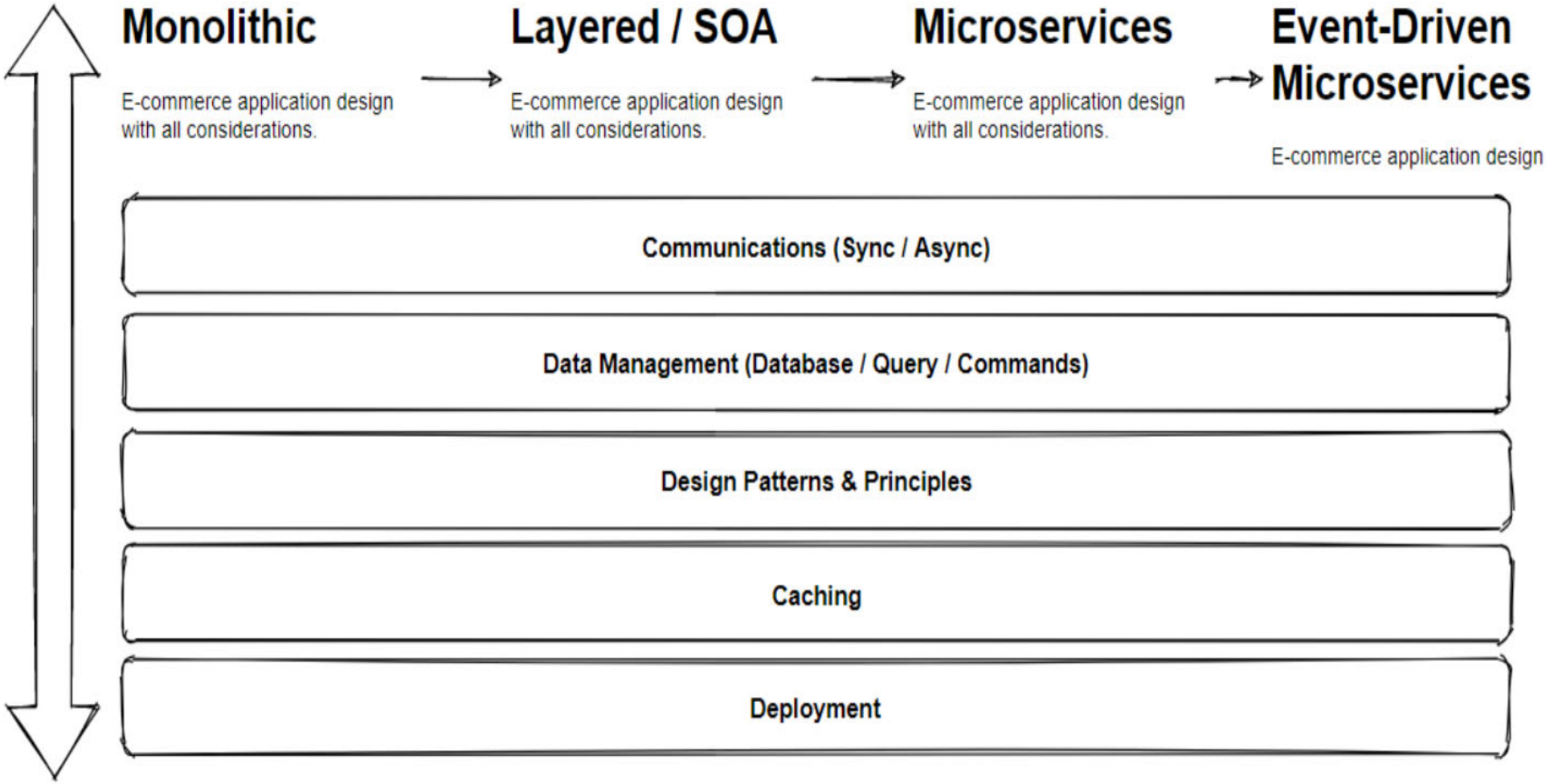
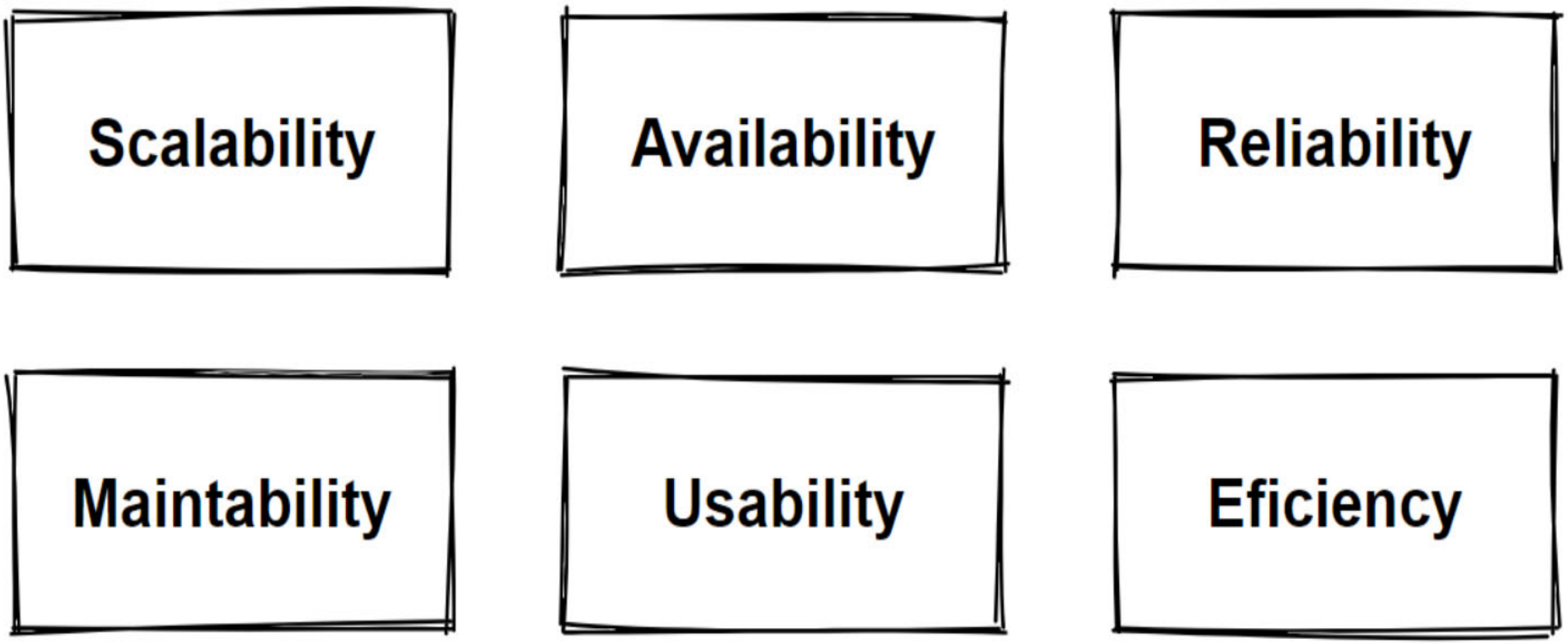
Lab3

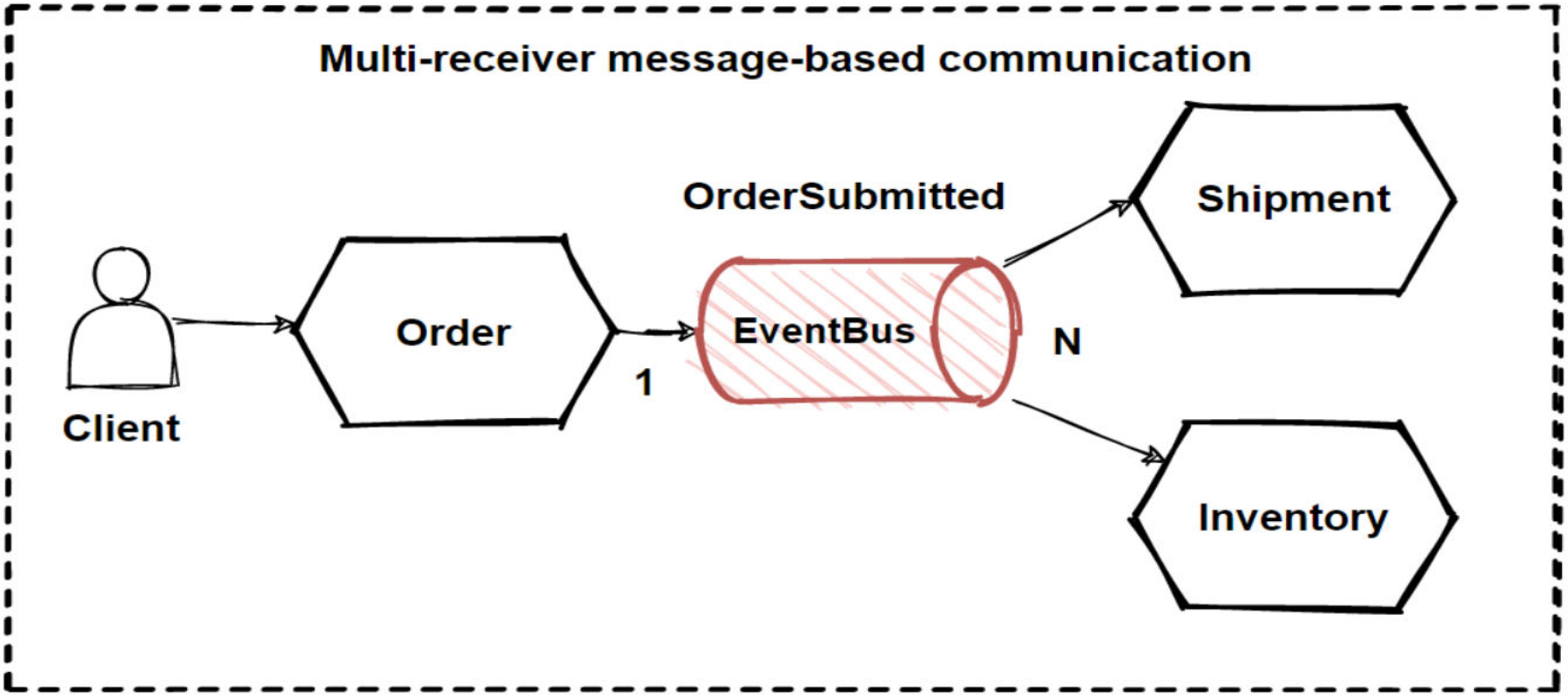
Архитектур хийх алхмууд

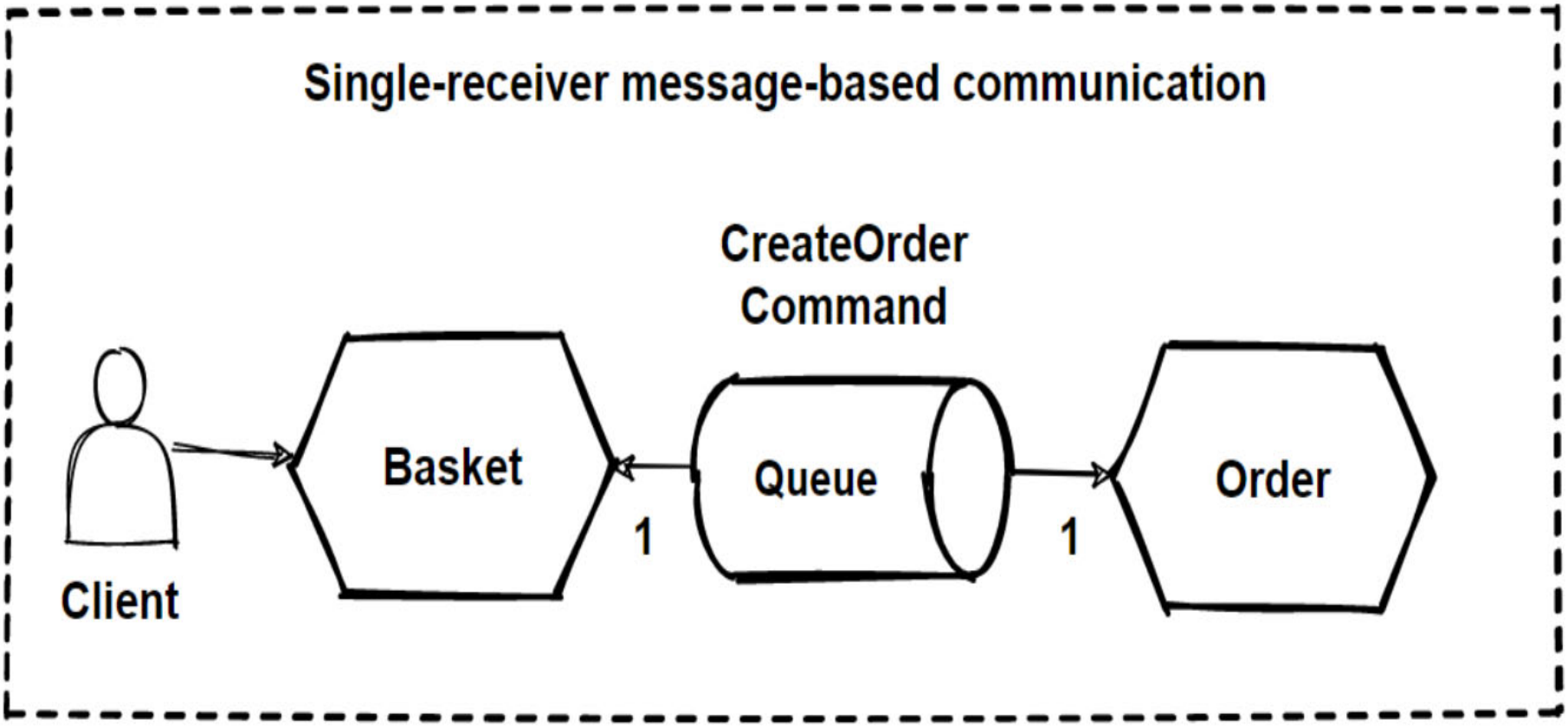



Monolithic
E-commerce application design with all considerations.
Layered / SOA
E-commerce application design with all considerations.
Microservices
E-commerce application design with all considerations.
Event-Driven Microservices
E-commerce application design
Communications (Sync / Async)
Data Management (Database / Query / Commands)
Design Patterns & Principles
Caching
Deployment

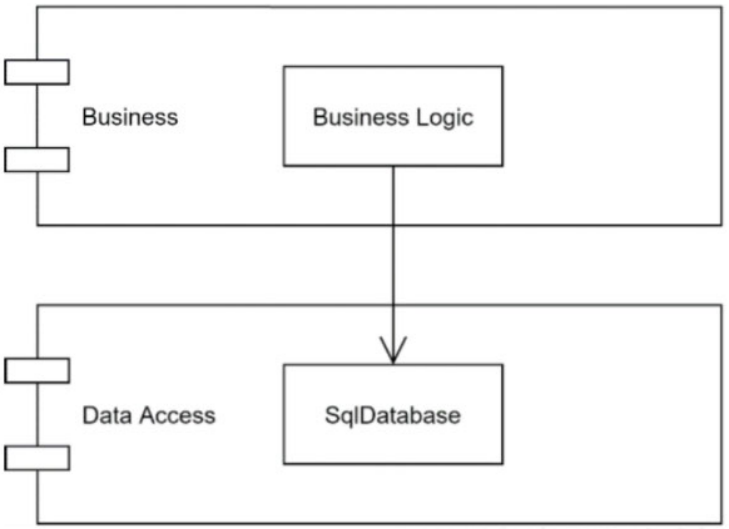
Non functional шаардлага



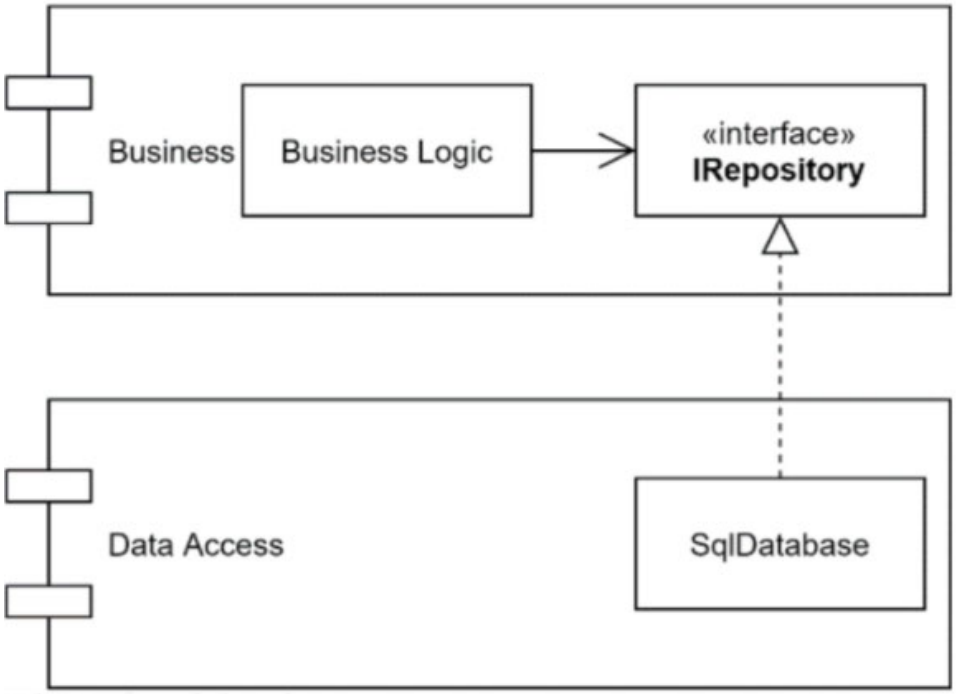

Multi-receiver message-based communication
OrderSubmitted
Shipment
Client
Order
1
EventBus
N
Inventory

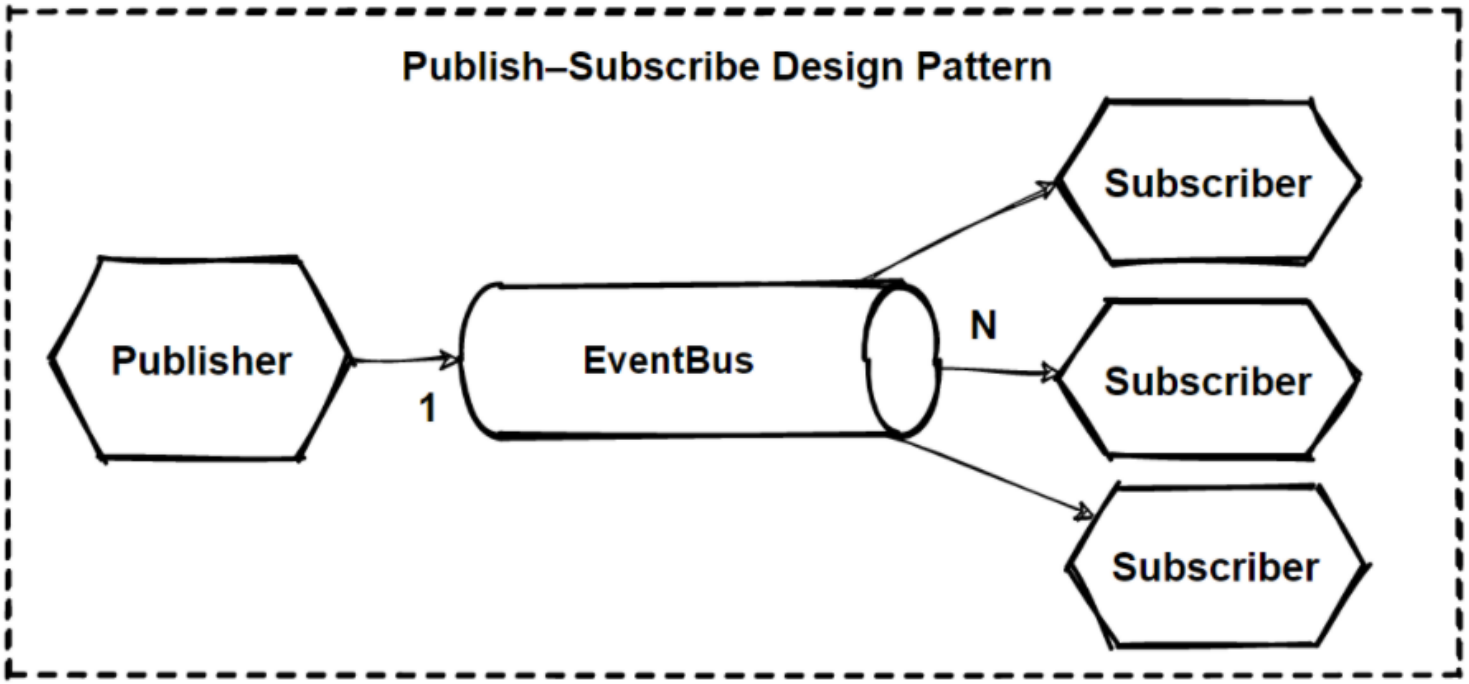

Single-receiver message-based communication
CreateOrder
Command
Basket
Queue
Order
1
1
Client

Without Dependency Inversion



With Dependency Inversion

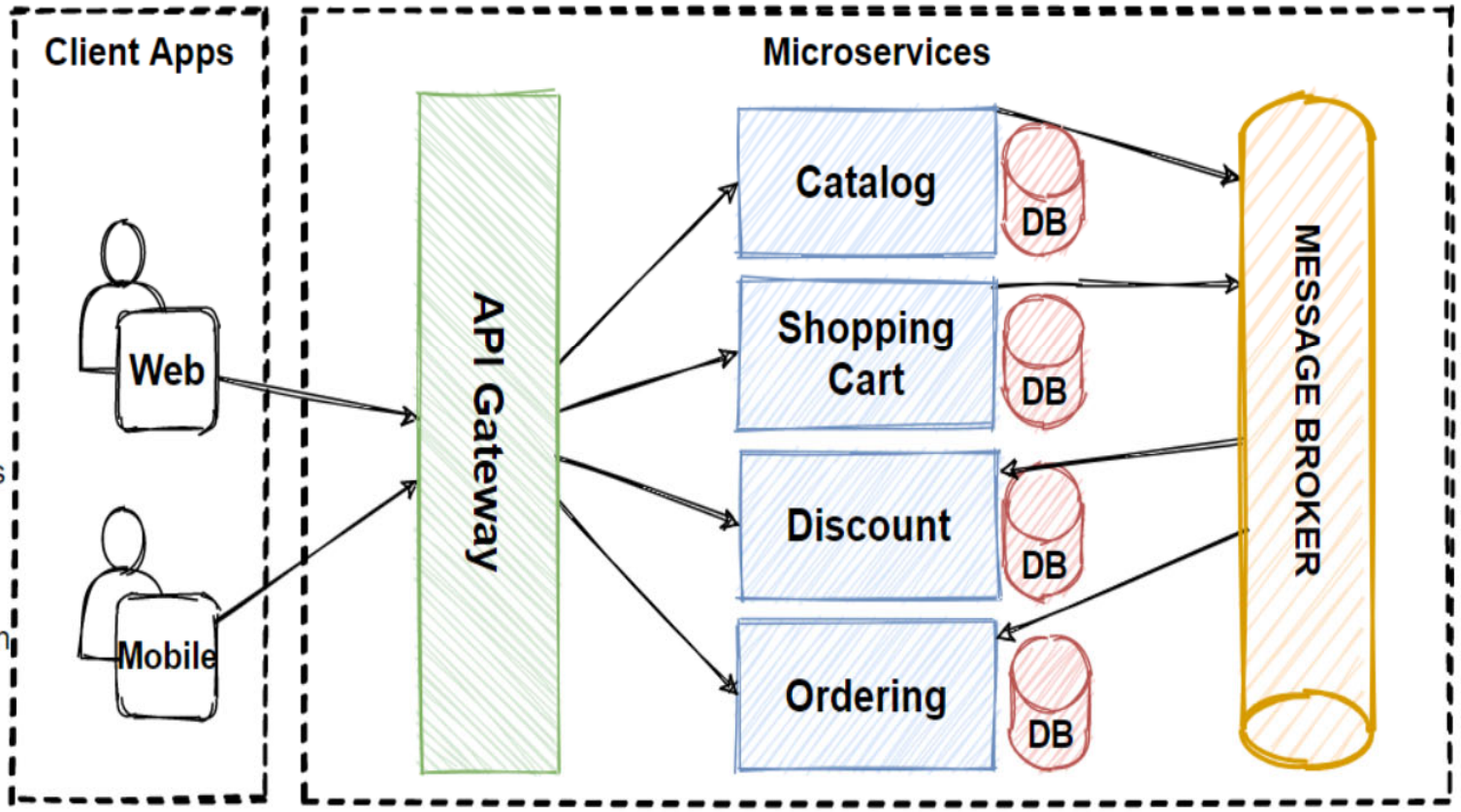


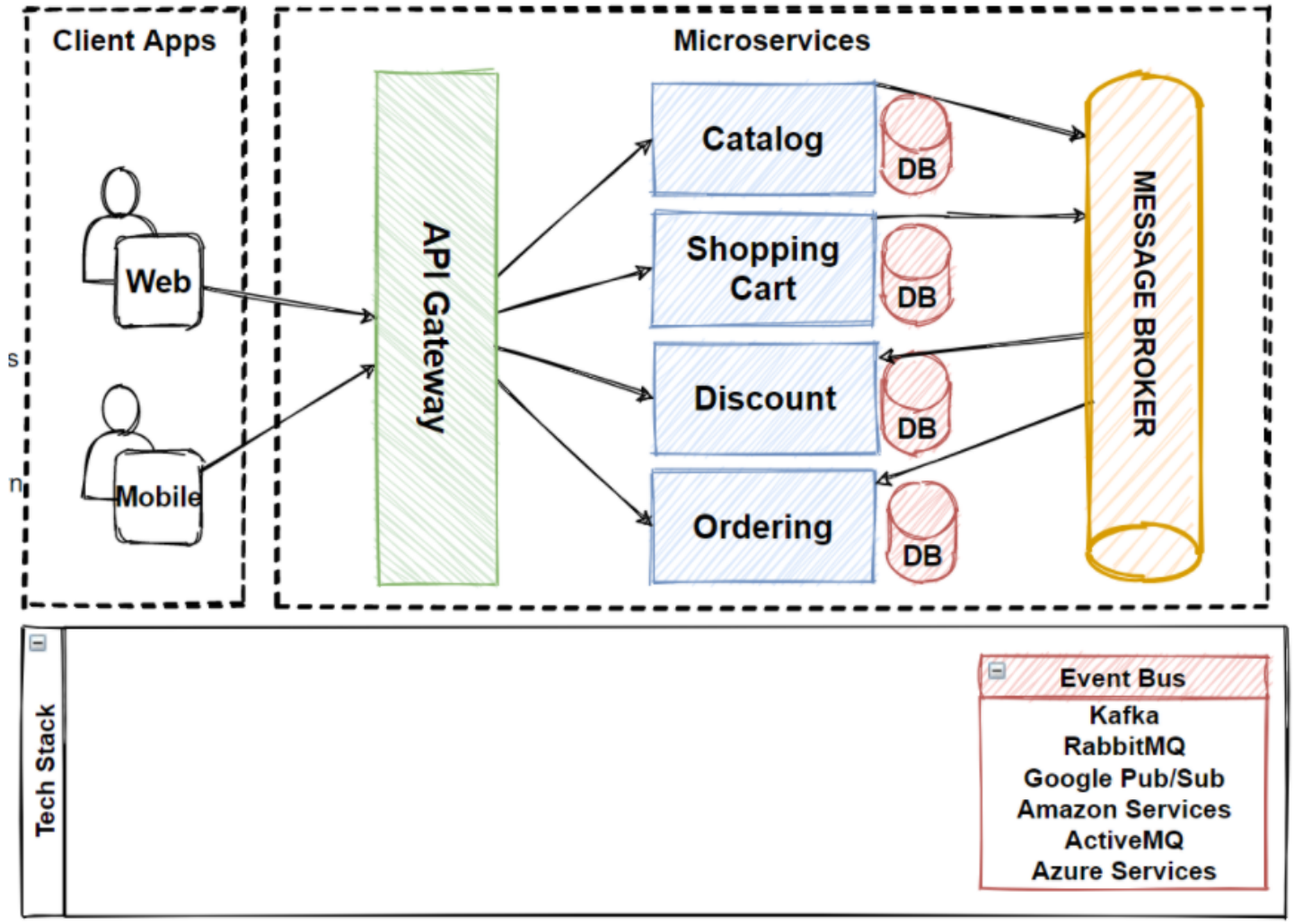

Publish–Subscribe Design Pattern
Publisher
1
EventBus
N
Subscriber
Subscriber
Subscriber

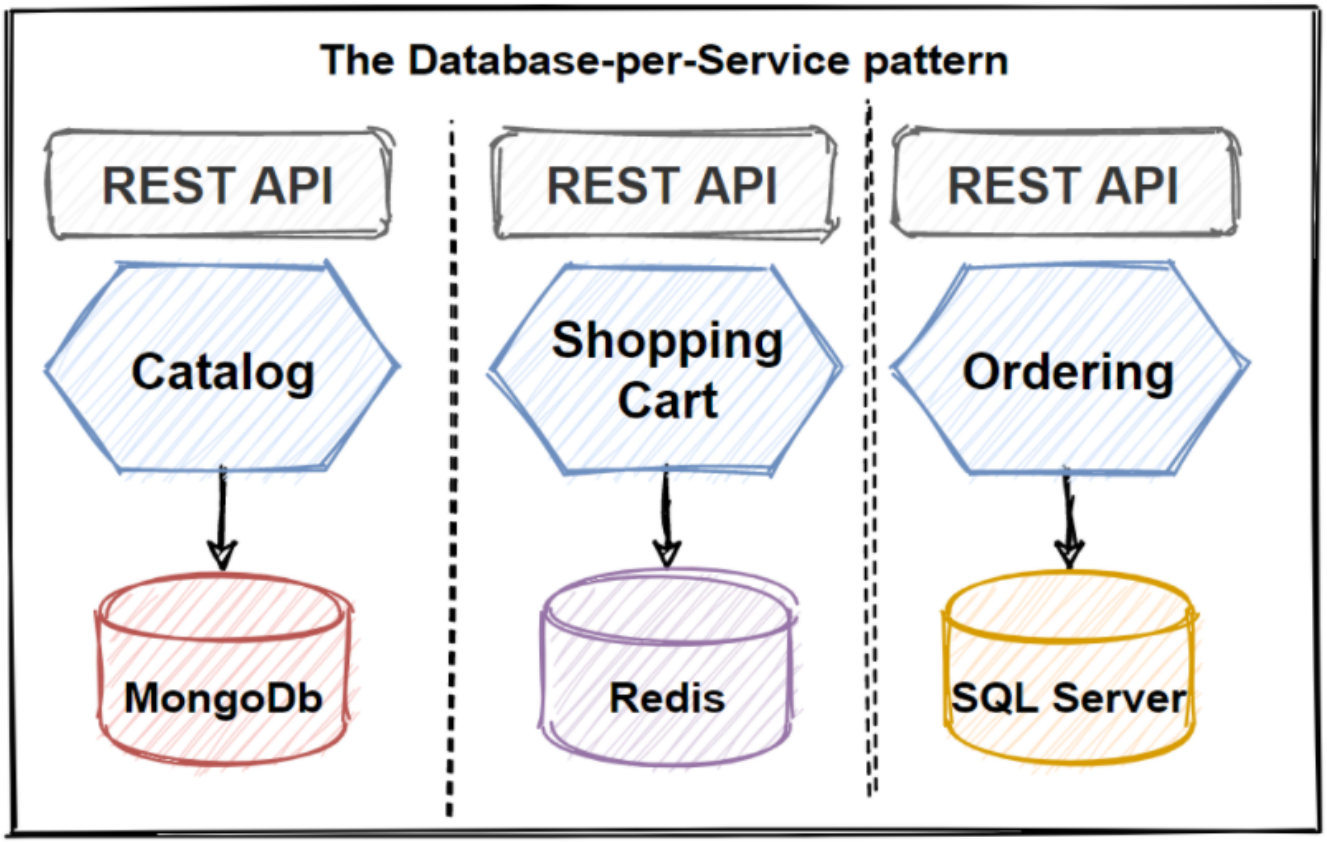
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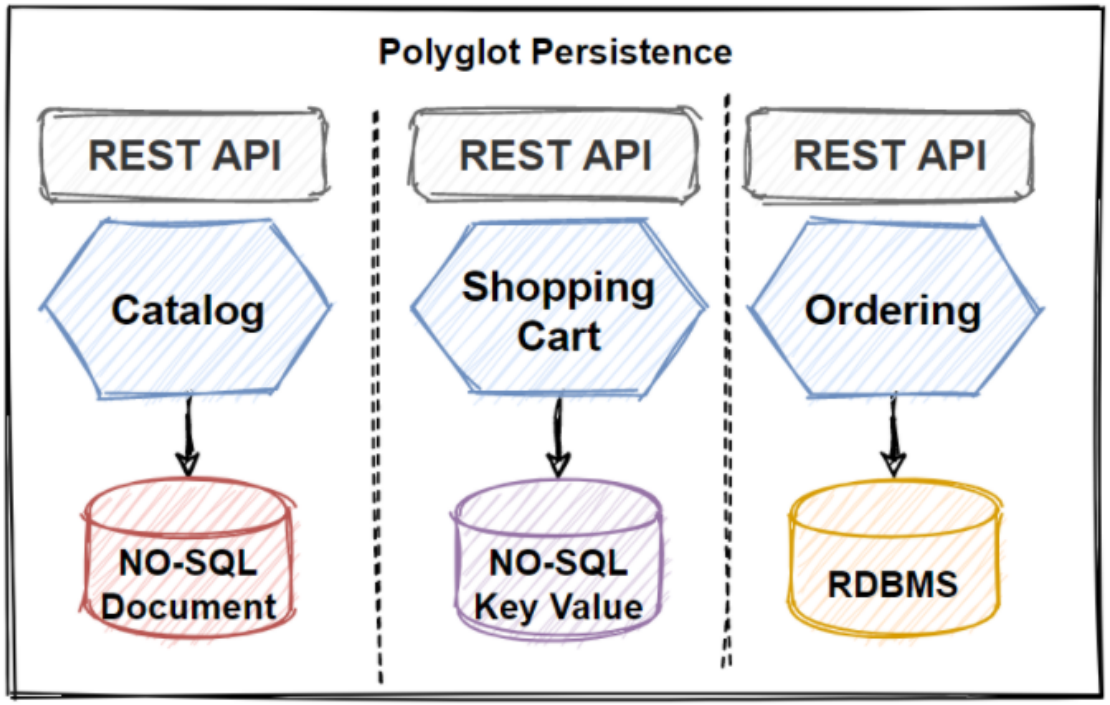
Principles

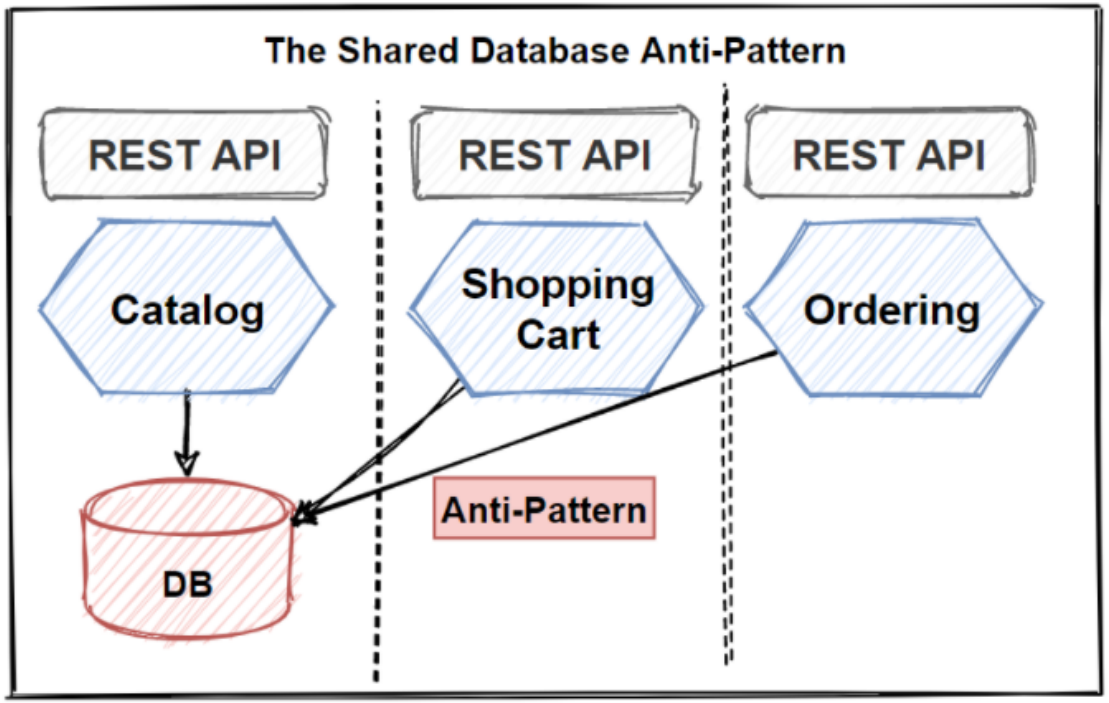
- KISS
- YAGNI
- SoC
- SOLID
- Gateway Routing
- Gateway Aggregation
- Gateway Offloading
- Api Gw
- Database per Microservices
- Backends for Frontends pattern BFF
- Service Aggregator Pattern
- Service Registry Pattern
- Dependency Inversion Principles (DIP)
- Publish–Subscribe Design Pattern




Client Apps
Microservices
Web
Mobile
API Gateway
Catalog
Shopping Cart
Discount
Ordering
DB
DB
DB
DB
MESSAGE BROKER
Tech Stack
Event Bus
Kafka
RabbitMQ
Google Pub/Sub
Amazon Services
ActiveMQ
Azure Services


The Database-per-Service pattern
REST API
REST API
REST API
Catalog
Shopping Cart
Ordering
MongoDb
Redis
SQL Server

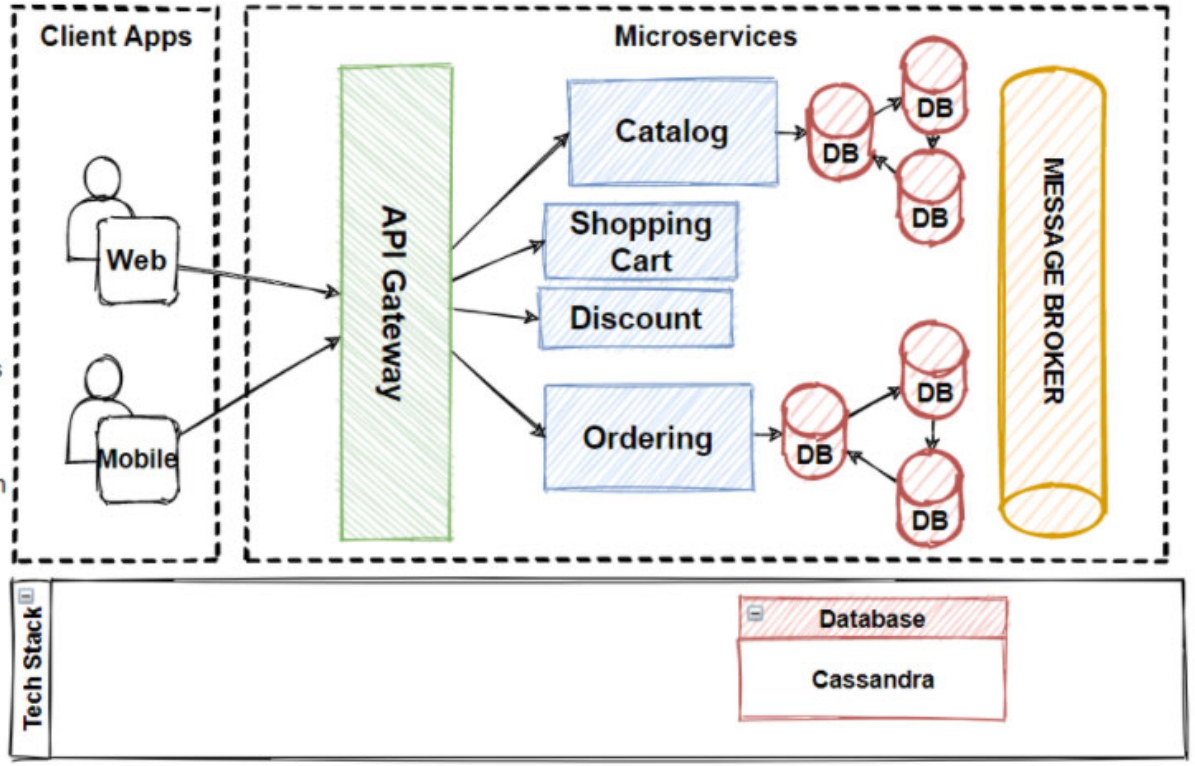

Polyglot Persistence
REST API
REST API
REST API
Catalog
Shopping Cart
Ordering
NO-SQL Document
NO-SQL Key Value
RDBMS

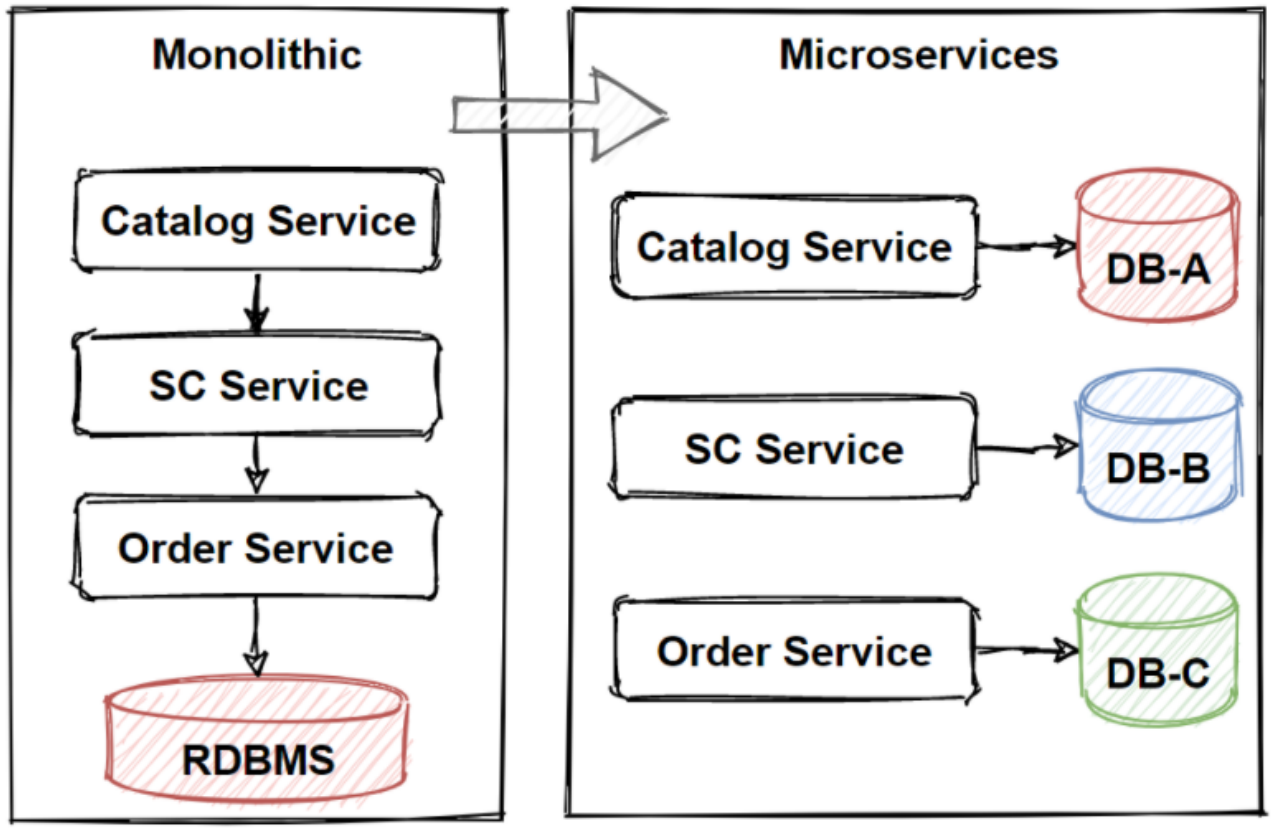

The Shared Database Anti-Pattern
REST API
REST API
REST API
Catalog
Shopping Cart
Ordering
Anti-Pattern
DB

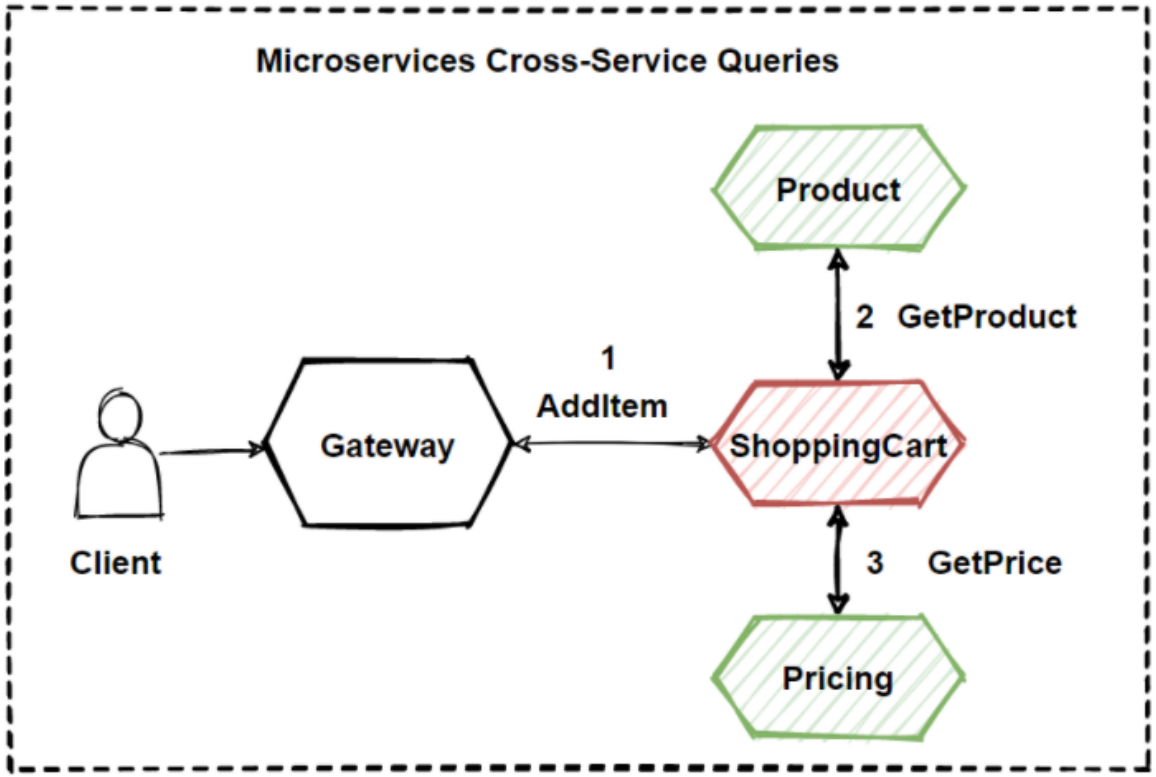
Microservices Architecture - Database Sharding

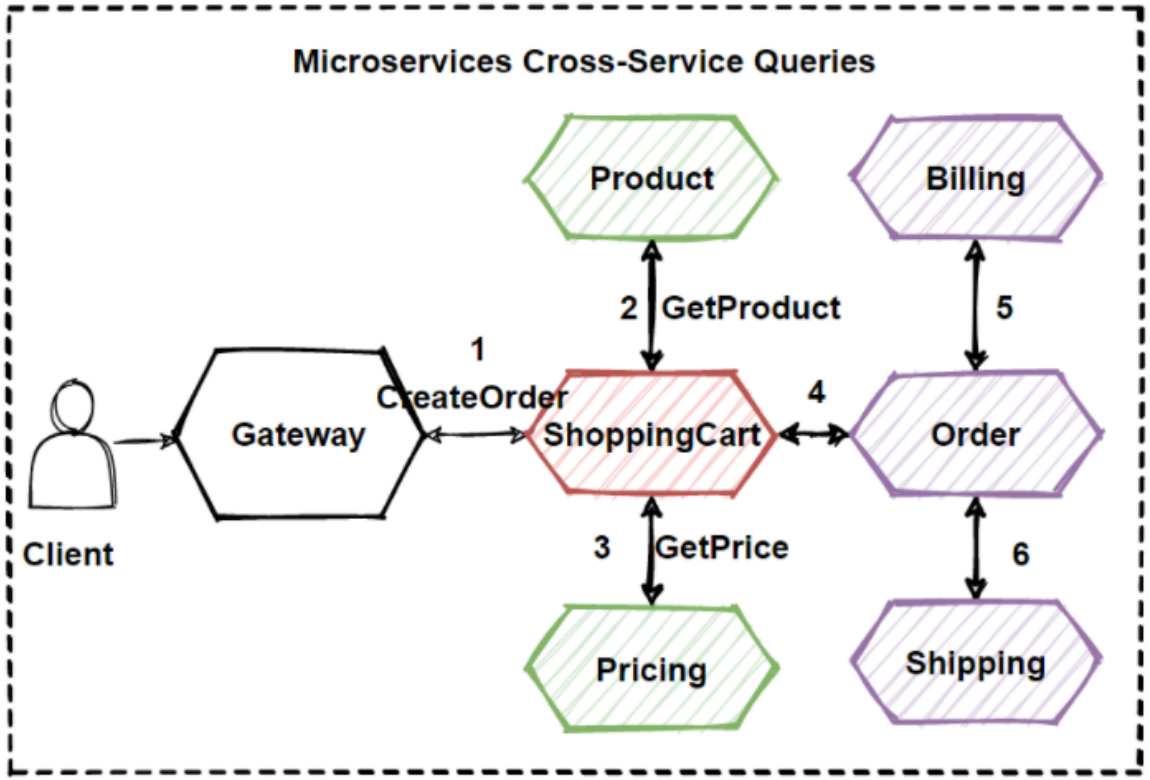
Principles

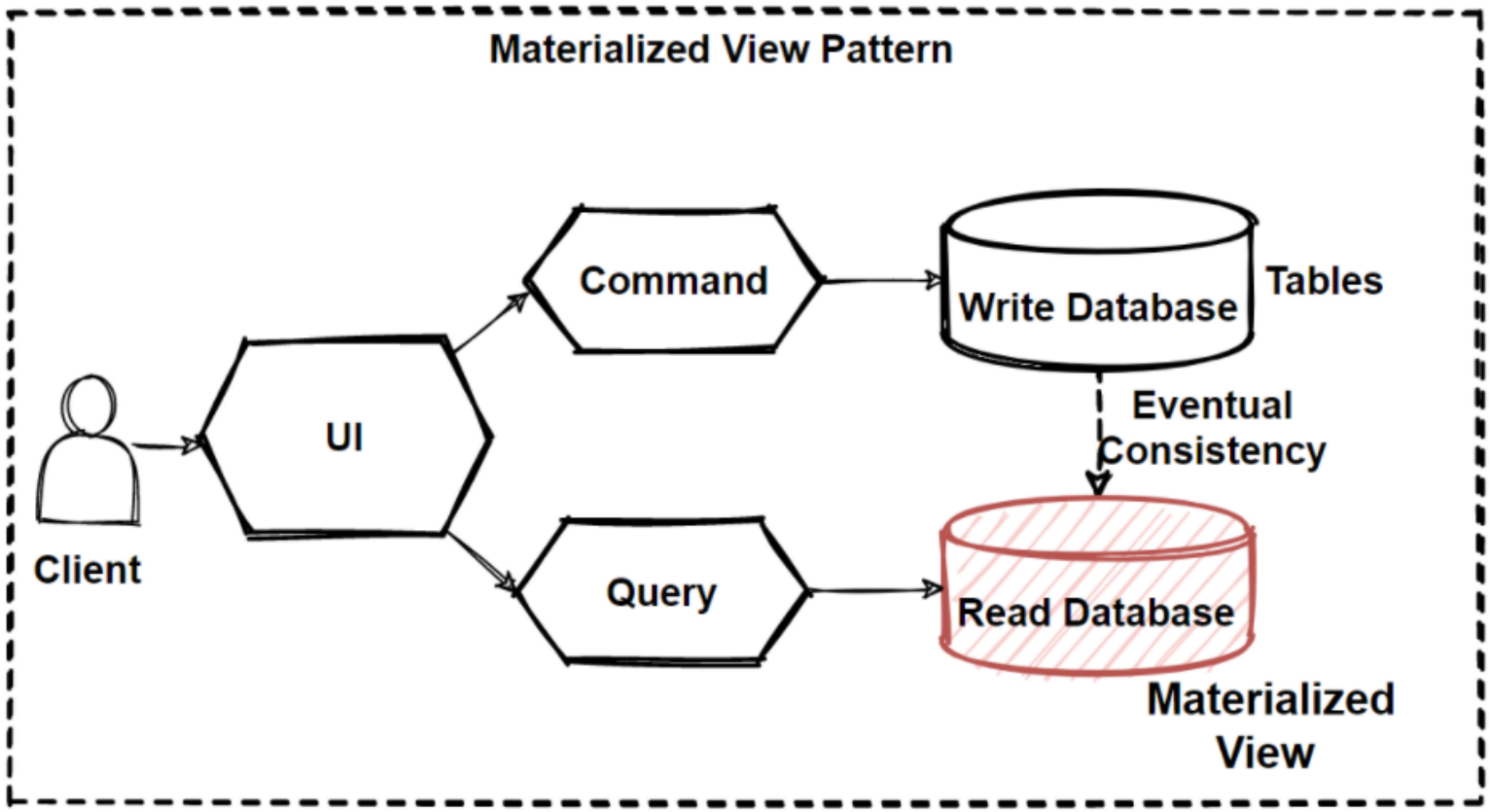
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- Publish–Subscribe Design Pattern
- Database Sharding Pattern

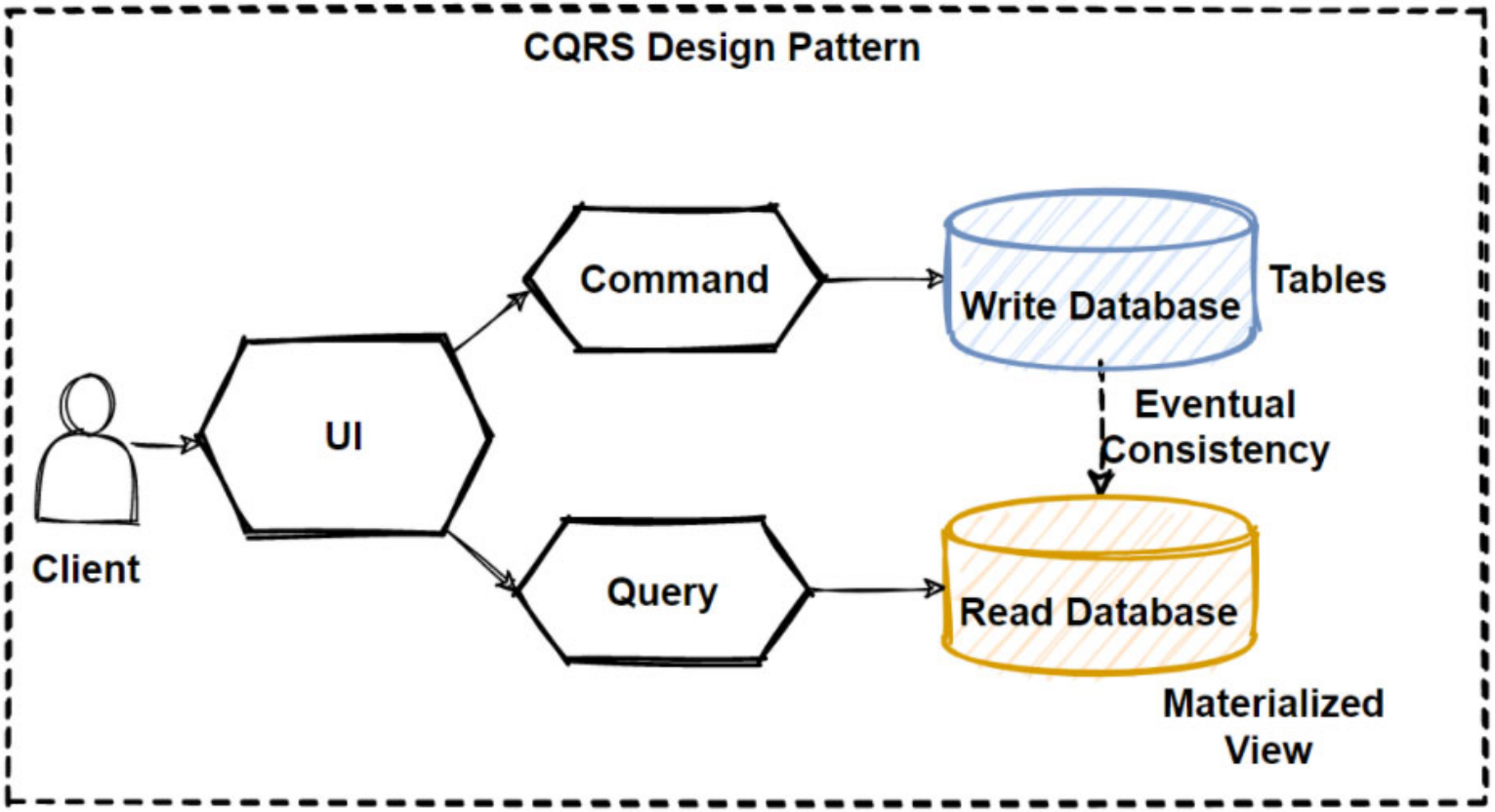


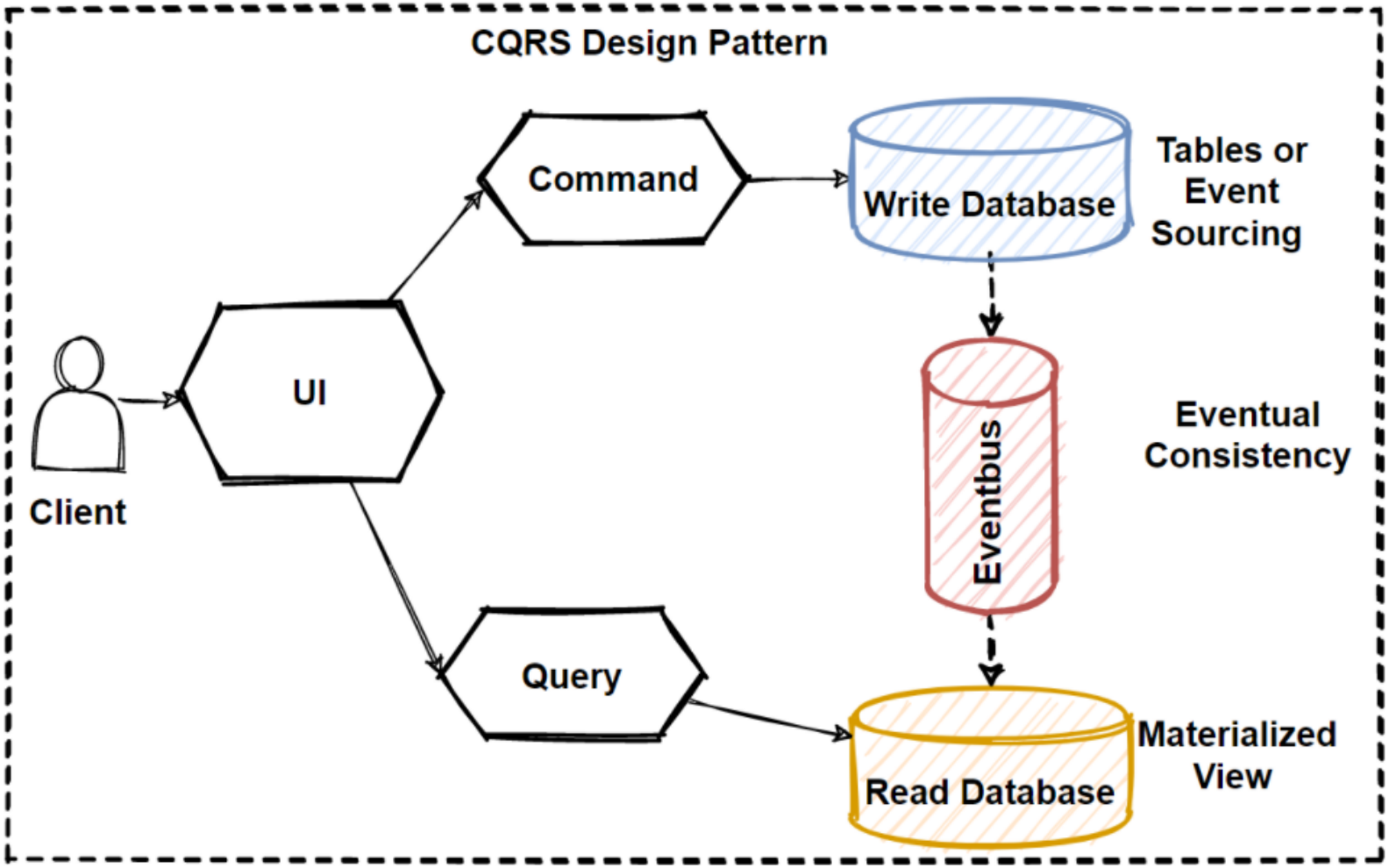

Monolithic
Catalog Service
SC Service
Order Service
RDBMS
Microservices
Catalog Service
DB-A
SC Service
DB-B
Order Service
DB-C

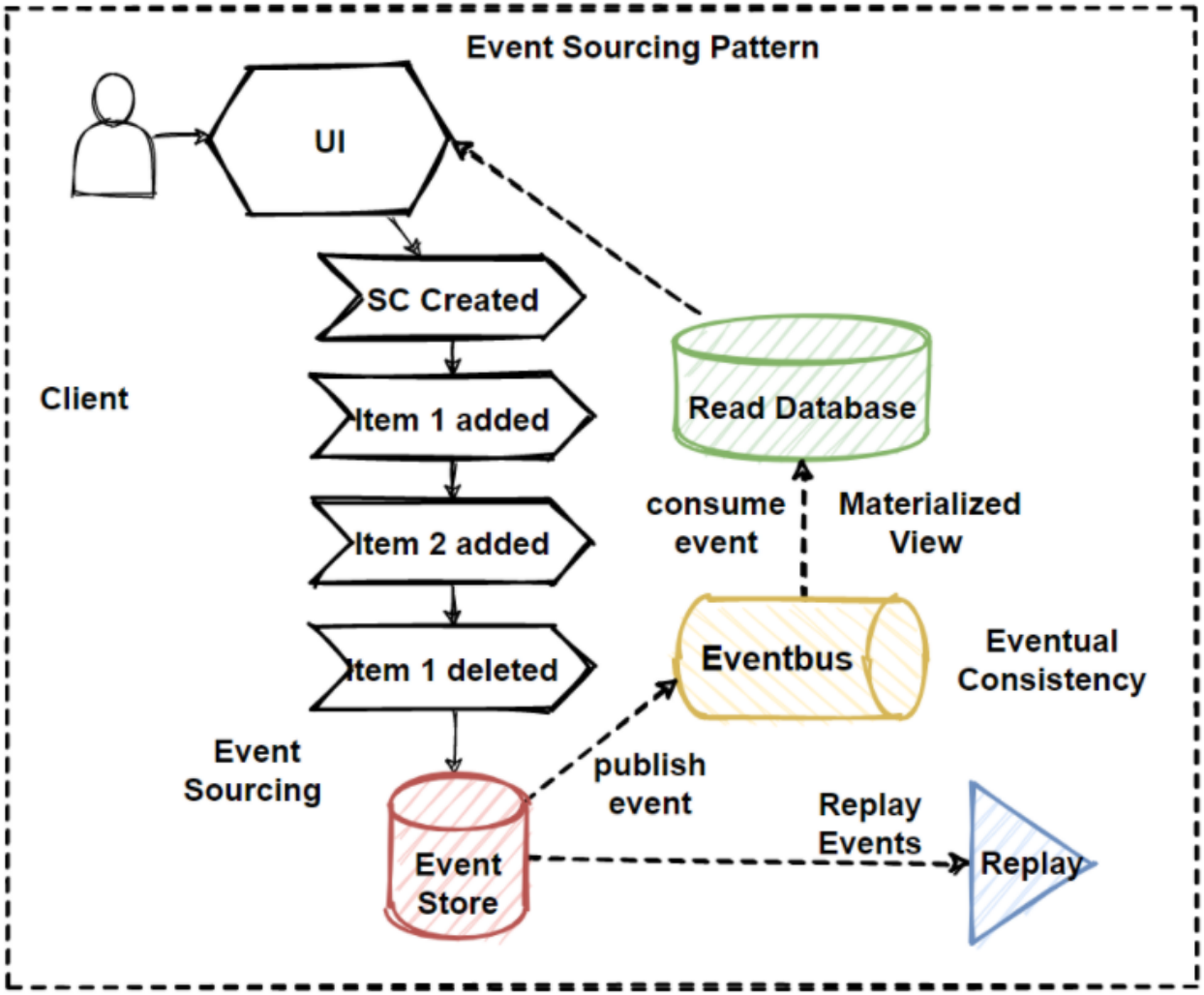

Microservices Cross-Service Queries
Product
2 GetProduct
1
AddItem
Gateway
ShoppingCart
Client
3 GetPrice
Pricing


Microservices Cross-Service Queries
Product
Billing
2
GetProduct
5
1
CreateOrder
4
Client
Gateway
ShoppingCart
Order
3
GetPrice
6
Pricing
Shipping


Materialized View Pattern
Command
Write Database
Tables
UI
Eventual
Consistency
Client
Query
Read Database
Materialized
View


CQRS Design Pattern
Command
Write Database
Tables
UI
Eventual
Consistency
Client
Query
Read Database
Materialized
View

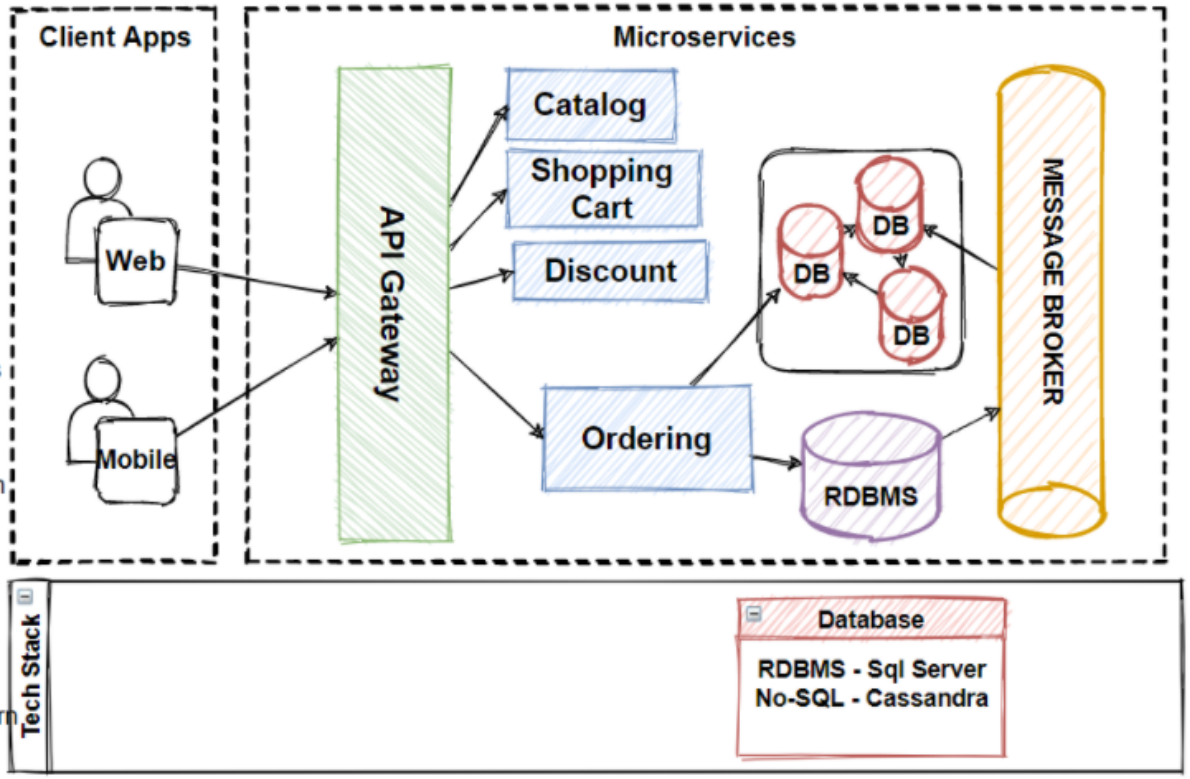

CQRS Design Pattern
Command
Write Database
Tables or
Event
Sourcing
UI
Client
Eventbus
Eventual
Consistency
Query
Read Database
Materialized
View

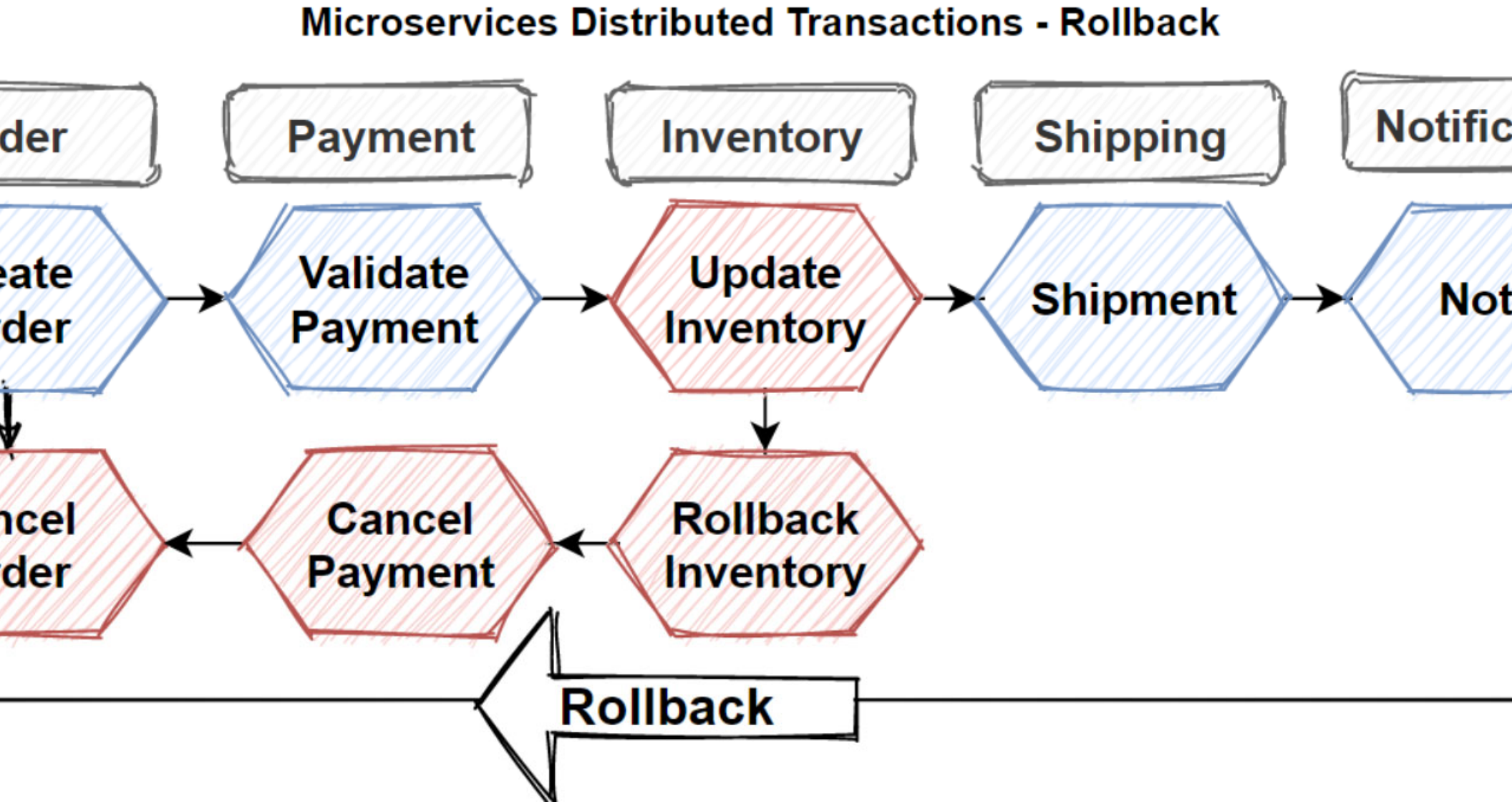

Event Sourcing Pattern
UI
SC Created
Item 1 added
Item 2 added
Item 1 deleted
Client
Read Database
consume event
Materialized View
Eventbus
Eventual Consistency
Event Sourcing
publish event
Replay Events
Event Store
Replay

Microservices Architecture - CQRS, Event Sourcing, Eventual Consistency

Principles

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- Polygot Persistence
- Materialized View Pattern
- CQRS Design Pattern
- Event Sourcing Pattern
- Eventual Consistency Principle




Microservices Distributed Transactions - Rollback
der
Payment
Inventory
Shipping
Notific
eate
rder
Validate
Payment
Update
Inventory
Shipment
Not
ncel
rder
Cancel
Payment
Rollback
Inventory
Rollback